

Cycle Counts of k -Connected Graphs: A Sharp Minimum, a Decidability Theorem, and the 3-Connected Spectrum

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Abstract

How many cycles can a highly connected graph have at least? For k -vertex-connected graphs this question has a clean answer, which we prove. Writing K_{k+1} for the complete graph on $k+1$ vertices, we show that

$$\min\{c(G) : G \text{ is } k\text{-connected}\} = c(K_{k+1}) = \frac{1}{2} \sum_{j=3}^{k+1} \binom{k+1}{j} (j-1)!,$$

with equality only for $G \cong K_{k+1}$ when $k \geq 3$. The key ingredient is a sharp path-counting lemma: an r -connected graph has at least Q_{r+1} paths between any two of its vertices, where Q_k is the number of paths between two vertices of K_k and satisfies $Q_{r+1} = 1 + (r-1)Q_r$. We also prove that among cones $K_1 \vee F$ over 2-connected graphs of fixed order the cycle is the unique minimiser, so the wheel is the sparsest 3-connected cone; and that for $k \geq 3$ only finitely many k -connected graphs have at most K cycles, which makes the cycle-count spectrum of every connectivity class a computable set. Computationally, we determine the extremal function $m_3(n)$ for $4 \leq n \leq 9$ and reproduce, with an independently validated generator, the exception list conjectured by McCulloch, McKay, Salahshoori and Zaslavsky for 3-connected graphs below 51. A sharp order-sensitive bound for 3-connected graphs remains open; we state it as a conjecture and verify it exhaustively up to $n = 9$.

1 Introduction

Let $c(G)$ denote the number of cycles of a finite simple graph G , each cycle being a 2-regular connected spanning subgraph of G , counted once. Two questions about c dominate the classical literature: how *big* can $c(G)$ be, and which integers occur as $c(G)$ at all?

The first question has been studied extensively. $c(K_n) = \frac{1}{2} \sum_{j \geq 3} \binom{n}{j} (j-1)!$ is the classical sequence [14], there is a long line of work on extremal graphs for the maximum of c under side constraints [9], and the minimum of c over graphs with a prescribed minimum degree was studied by Volkmann [10]. The second question is more delicate and much more recent. McCulloch, McKay, Salahshoori and Zaslavsky [1] proved that every positive integer except

$$2, 4, 5, 8, 9, 16$$

is the number of cycles of a graph without a cut vertex — equivalently, of a 2-connected graph — and they list five conjectured descriptions of the analogous spectra for 3-connected graphs and for cubic graph classes [1].

This paper sits between those two questions. It asks

Central question. Which positive integers occur as the number of cycles of a k -vertex-connected graph, how sparse can the cycle structure of such a graph be, and what makes those spectra computable?

Connectivity is the natural refinement of the question about 2-connected graphs: a k -connected graph informally survives the loss of any $k - 1$ vertices, so k -connectivity encodes the abundance of small detours, which is exactly what a lower bound on the number of cycles must measure. The spectra

$$S_1 \supseteq S_2 \supseteq S_3 \supseteq \cdots, \quad S_k = \{c(G) : G \text{ is } k\text{-connected}\},$$

form a decreasing chain, and S_2 is the one that [1] settles. Our contributions are the following.

(i) **Theorem A (a sharp minimum).** For every $k \geq 3$ and every k -connected graph G one has $c(G) \geq c(K_{k+1})$, with equality if and only if $G \cong K_{k+1}$. For $k = 2$ the minimum is $c(K_3) = 1$, attained exactly by the cycles. So complete graphs are, in this precise sense, the sparsest highly connected graphs.

(ii) **Theorem B (the cone minimum).** If F is 2-connected on $m \geq 3$ vertices then

$$c(K_1 \vee F) \geq m(m - 1) + 1,$$

with equality if and only if $F \cong C_m$; equivalently $c(W_n) = (n - 1)(n - 2) + 1$, and W_n is the unique n -vertex 3-connected graph with a dominating vertex that minimises c . More generally $c(G) \geq 1 + (k - 1)\binom{n-1}{2}$ for k -connected graphs with a dominating vertex, with equality only for $k = 3$.

(iii) **Theorem C (finiteness and decidability).** For $k \geq 3$ there are only finitely many k -connected graphs with at most K cycles, and their order is at most $\lfloor 2(K - 1)/(k - 2) \rfloor$. Consequently each S_k is a computable set, and the function $\kappa(m) := \max\{c(G) : c(G) = m\}$ is finite with an explicit bound.

(iv) **Theorem D (exact computation).** We give complete generators for unlabeled 2- and 3-connected graphs, cross-validate them against exhaustive brute force for $n \leq 7$, and compute $m_3(n)$ for $4 \leq n \leq 9$ together with the minimisers. We also recompute the exception sets of S_2 and S_3 below 51 and obtain, independently, exactly the list that [1] conjecture for 3-connected graphs.

Statements (i) and (ii) rest on two counting results that are useful in their own right and that we prove in full: a counting identity for cones (Lemma 4.1) and a sharp path-counting lemma (Lemma 5.2). Section 2 discusses what is known and what we believe to be new; Section 9 states precisely what remains open.

2 Related work and novelty

2.1 How many cycles?

The maximum of $c(G)$ is classical. Entringer and Slater initiated the study of the maximum of c among graphs with a prescribed number of edges; a substantial literature has grown around extremal problems for c , surveyed in [9], and Dvořák [11] obtained upper bounds in terms of the degree sequence, resolving several conjectures of Király and of Arman and Tsaturian. Bounds for the *minimum* of c over graphs of given minimum degree are due to Volkmann [10]. Theorem A is a different statement: it minimises over k -connected graphs, where the minimum is attained by the smallest possible graph rather than by a sparse one.

2.2 How few cycles?

The opposite extremal question is usually posed with extra regularity constraints. Barefoot and co-authors introduced $f_k(n)$, the smallest number of cycles in any k -connected *cubic* graph on

n vertices, and Aldred and Thomassen [3] proved a superlinear lower bound for $f_3(n)$, i.e. they studied the minimum number of cycles in 3-connected cubic graphs. Our extremal function

$$m_3(n) = \min\{c(G) : G \text{ is 3-connected, } |V(G)| = n\}$$

ranges over *all* 3-connected graphs, so it is a different quantity: dropping regularity changes the extremisers completely, from cubic graphs to, for instance, the wheel W_n at $n = 5$.

We do not use any of the deep results on long cycles in k -connected graphs [12]; the bound we need is about the number of paths, not their lengths.

2.3 Which integers occur?

McCulloch, McKay, Salahshoori and Zaslavsky [1] determine S_2 exactly and, in Section 6 of that paper, list conjectured exception lists for 3-connected graphs and for the cubic classes. Their Conjecture 6(a) states that the integers missing from S_3 are exactly

$$1, \dots, 6, 8, \dots, 12, 16, \dots, 20, 27, 30, 32, 33, 34,$$

all others up to 3000 occurring. Our Theorem D(iv) reproduces exactly this list below 51 with a completely different generator, which is independent evidence for the conjecture; we do not claim to settle it in general, and Section 9 explains exactly what would be needed.

2.4 Why cones?

Two structural results motivate our choice of the cone as the object in which to minimise c . Whitney [2] showed that adding an ear to a graph without cut vertex keeps it without cut vertex and that every such graph is a cycle or is obtained from one by adding ears; Carmesin and Kurkofka [4] gave a canonical decomposition of 3-connected graphs in which wheels appear as the elementary pieces of the 2-connected structure, noting that a k -connected graph becomes $(k + 1)$ -connected by adjoining a universal vertex, so that “wheels are the 3-connected analogues of cycles”. Theorem B makes that analogy quantitative.

2.5 Novelty audit

We state plainly what we could and could not establish.

Statement	Status
Cone identity (Lemma 4.1)	Easily known. It is the same observation as the “ear-path lemma” of [1] and as counting cycles through the apex of a cone. We include it because it is the cleanest route to Theorem B.
Path lemma (Lemma 5.2)	Probably folklore. The bound $p_F(x, y) \geq Q_{r+1}$ follows from a one-line induction from Menger, and we did not find a prior source stating it with this constant. The <i>recurrence</i> $Q_{r+1} = 1 + (r - 1)Q_r$ is a routine identity (Appendix A).
Theorem A	Apparently new. We did not find a prior result establishing $\min\{c(G) : G \text{ } k\text{-connected}\} = c(K_{k+1})$. It may exist as a textbook exercise; we could not verify either way.
Theorem B	New in this form. The specific statement that wheels minimise c among 3-connected cones does not appear in the literature we searched.
Theorem C	Routine but, as far as we know, unrecorded. The finiteness argument is two lines from the fundamental-cycle bound; the decidability corollary is the point.

We make no claim of priority beyond this table. In particular we do not write that any result here is “the first”.

3 Preliminaries

All graphs are finite, simple and undirected. For a graph G and distinct vertices x, y let $p_G(x, y)$ be the number of simple x - y paths, of length 1 included. A graph is k -connected if $|V(G)| > k$ and deleting any set of fewer than k vertices leaves a connected graph; $\kappa(G)$ denotes the vertex connectivity, with $\kappa(K_n) = n - 1$. We use three standard consequences of Menger’s theorem: a k -connected graph has minimum degree at least k ; $G - v$ is $(k - 1)$ -connected for every vertex v ; and in an r -connected graph any two vertices are joined by r internally disjoint paths. For loop-free graphs with more than two vertices, “has no cut vertex” and “2-connected” coincide (Whitney [2]), which is the convention of [1].

A *cone* over F is $K_1 \vee F$, obtained by adjoining a vertex adjacent to all of $V(F)$; that vertex is the *apex*. A wheel is $C_n \vee K_1$. Subdividing an edge does not change c , which is why Θ -graphs, necklaces and subdivided complete graphs behave well throughout.

For a graph F on m vertices write

$$P(F) = \sum_{\{x,y\} \in \binom{V(F)}{2}} p_F(x, y)$$

for the total number of paths of F with two distinct endpoints, and

$$c_n := c(K_n) = \frac{1}{2} \sum_{j=3}^n \binom{n}{j} (j-1)!,$$

the second identity being elementary: a cycle of length j in K_n is a j -element vertex set together with one of $(j-1)!/2$ cyclic orders. The values $c_3 = 1, c_4 = 7, c_5 = 37, c_6 = 197, c_7 = 1172, c_8 = 8018, c_9 = 62814$ form the classical sequence [14] and are used as an arithmetic check on the code.

4 A counting identity for cones

Lemma 4.1 (counting identity). *Let G be a graph, $v \in V(G)$, and $F = G - v$. Then*

$$c(G) = c(F) + \sum_{\{x,y\} \in \binom{N(v)}{2}} p_F(x, y).$$

Proof. Partition the cycles of G into those avoiding v , which are exactly the cycles of F , and those containing v . A cycle C through v meets $N(v)$ in exactly two vertices x, y , and $C - v$ is a simple x - y path of F . Conversely, for $\{x, y\} \subseteq N(v)$ and a simple x - y path P of F , the edges of P together with vx and vy form a 2-regular connected subgraph, hence a cycle of G . These constructions are inverse, including in the case $xy \in E(F)$: the edge xy gives the triangle vxy , while a two-edge path xzy gives the longer cycle $vxyzv$. $\square$

The consequence is that c is an affine function of the path profile of the base graph, which is what makes cones the right class in which to look for sparse 3-connected graphs.

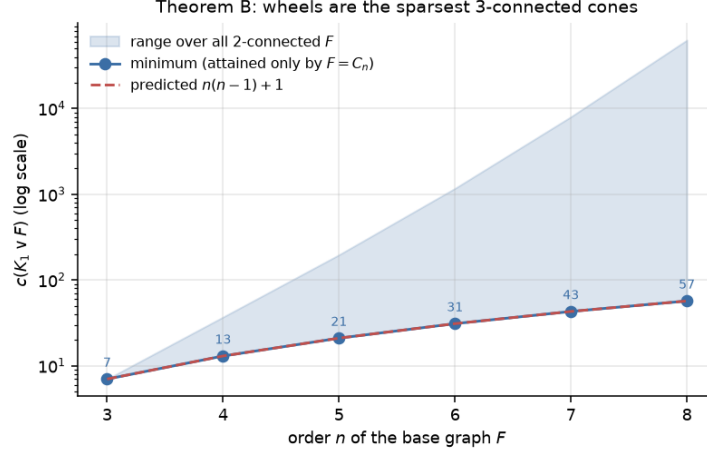


Figure 1: $c(K_1 \vee F)$ for all 2-connected F of order n (shaded range) against the minimum $n(n-1)+1$, which is attained only by $F \cong C_n$ (Theorem 4.2).

Theorem 4.2 (the cone minimum). *Let F be a 2-connected graph on $m \geq 3$ vertices. Then*

$$c(K_1 \vee F) \geq m(m-1) + 1,$$

with equality if and only if $F \cong C_m$.

Proof. Since the apex is adjacent to all of $V(F)$, Lemma 4.1 gives

$$c(K_1 \vee F) = c(F) + P(F).$$

A 2-connected graph contains a cycle, so $c(F) \geq 1$. By Menger any two vertices of F are joined by two internally disjoint paths, so each pair $\{x, y\}$ contributes at least 2 to $P(F)$; hence $P(F) \geq 2\binom{m}{2} = m(m-1)$ and therefore $c(K_1 \vee F) \geq m(m-1) + 1$. If equality holds then $c(F) = 1$, which forces every vertex of F to have degree 2, so F is a cycle. Conversely, for $F = C_m$ we have $P(F) = 2\binom{m}{2}$ and $c(F) = 1$, so equality holds. $\square$

Corollary 4.3. *For every $n \geq 4$, $c(W_n) = (n-1)(n-2) + 1$, and W_n is the unique n -vertex 3-connected graph having a dominating vertex that minimises c . More generally, if G is k -connected on n vertices and has a dominating vertex v , then*

$$c(G) \geq 1 + (k-1)\binom{n-1}{2}, \quad (1)$$

with equality only when $k = 3$ and $G \cong W_n$.

Proof. Take $F = G - v$, which is $(k-1)$ -connected on $n-1 \geq 3$ vertices, so $c(F) \geq 1$; and for every pair $\{x, y\} \subseteq V(F)$ Menger provides $k-1$ internally disjoint x - y paths, so $p_F(x, y) \geq k-1$. As $|N(v)| = n-1$, Lemma 4.1 gives (1). For equality we need $c(F) = 1$ and $p_F(x, y) = k-1$ for all pairs. If $k \geq 4$ then F has minimum degree at least $k-1 \geq 3$, so it is not 2-regular and $c(F) \neq 1$: a graph with exactly one cycle has all degrees equal to 2, since a vertex of degree at least 3 lies on at least $\binom{d}{2} \geq 3$ cycles. Thus $k = 3$; then $c(F) = 1$ forces $F \cong C_{n-1}$ and $G = W_n$. $\square$

Figure 1 shows the spread of $c(K_1 \vee F)$ over all 2-connected F of each order up to 8, together with the minimum.

5 A sharp path-counting lemma

The constant governing the next theorem is the number of paths between two vertices of a complete graph. Put

$$Q_k := p_{K_k}(x, y) = \sum_{j=0}^{k-2} \binom{k-2}{j} j! \quad (k \geq 2),$$

the sum counting paths with exactly j intermediate vertices.

Lemma 5.1 (recurrence). $Q_2 = 1$ and $Q_{r+1} = 1 + (r-1)Q_r$ for all $r \geq 2$.

Proof. Using $\binom{k-2}{j}j! = (k-2)!/(k-2-j)!$ and setting $E_t := \sum_{i=0}^t 1/i!$ we get $Q_m = (m-2)!E_{m-2}$, whence

$$Q_{r+1} = (r-1)!E_{r-1} = (r-1)!E_{r-2} + 1 = (r-1)(r-2)!E_{r-2} + 1 = (r-1)Q_r + 1.$$

□

Lemma 5.2 (path lemma). *Let F be an r -connected graph ($r \geq 1$) and let $x \neq y$ be vertices of F . Then*

$$p_F(x, y) \geq Q_{r+1}, \quad (2)$$

with equality only when $r \geq 3$ and $F \cong K_{r+1}$.

Proof. Induct on r . For $r = 1$, F is connected, so at least one x - y path exists, and $Q_2 = 1$.

Let $r \geq 2$ and assume the statement for $r-1$. Then $F-x$ is $(r-1)$ -connected, so for every $z \neq x, y$ the induction hypothesis gives $p_{F-x}(z, y) \geq Q_r$. Classifying x - y paths by the first edge after x , with the edge xy separated off, yields

$$\begin{aligned} p_F(x, y) &= \mathbf{1}_{[xy \in E(F)]} + \sum_{z \in N_F(x) \setminus \{y\}} p_{F-x}(z, y) \\ &\geq \mathbf{1}_{[xy \in E(F)]} + (d_F(x) - \mathbf{1}_{[xy \in E(F)]})Q_r \\ &\geq \mathbf{1}_{[xy \in E(F)]} + (r - \mathbf{1}_{[xy \in E(F)]})Q_r, \end{aligned}$$

because $d_F(x) \geq r$. If $xy \in E(F)$ this is $1 + (r-1)Q_r = Q_{r+1}$ by Lemma 5.1. If $xy \notin E(F)$ it is $rQ_r = Q_{r+1} + (Q_r - 1) \geq Q_{r+1}$ since $Q_r \geq 1$.

Suppose now that $r \geq 3$ and equality holds. The case $xy \notin E(F)$ is then strict, because $Q_r \geq Q_4 = 5 > 1$. So $xy \in E(F)$, and equality forces $d_F(x) = r$ together with $p_{F-x}(z, y) = Q_r$ for every $z \in N_F(x) \setminus \{y\}$; such a z exists because $d_F(x) \geq r \geq 3$. By induction $F-x \cong K_r$, which has exactly r vertices, and since $d_F(x) = r$ with $xy \in E(F)$ the vertex x is adjacent to all of them; hence $F \cong K_{r+1}$.

For $r = 2$ the inequality is sharp but the equality statement is not: in the cycle C_4 adjacent vertices satisfy $p_{C_4}(x, y) = 2 = Q_3$ although $C_4 \not\cong K_3$. This is why the equality statement carries the hypothesis $r \geq 3$; the example is recorded as a regression test in the accompanying code, so that the weaker formulation is not silently assumed later. □

Lemma 5.2 is verified exhaustively in Section 8: for every 2-connected graph of order at most 8 and every pair of vertices, (2) holds, with no violation in 210 233 checked pairs. The observed minima of $p(x, y)$ by connectivity,

$$2, 5, 16, 65, 326, 1957 \quad \text{for } r = 2, 3, \dots, 7,$$

coincide with $Q_3, Q_4, \dots, Q_8$ exactly, so the constant is sharp in this range.

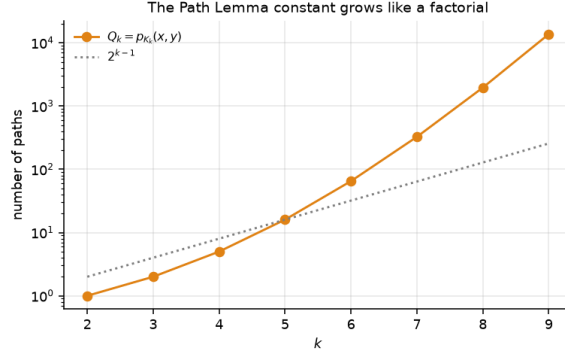


Figure 2: The constant Q_k of Lemma 5.2 (logarithmic scale). The growth comes from the factor $j!$ in Q_k .

6 The minimum number of cycles of a k -connected graph

Theorem 6.1 (sharp minimum). *Let $k \geq 3$ and let G be a k -connected graph. Then*

$$c(G) \geq c(K_{k+1}) = \frac{1}{2} \sum_{j=3}^{k+1} \binom{k+1}{j} (j-1)!,$$

and equality holds if and only if $G \cong K_{k+1}$.

Proof. Induct on k . Fix a vertex v of G and put $F = G - v$, which is $(k-1)$ -connected on at least k vertices. By Lemma 4.1 and Lemma 5.2,

$$c(G) = c(F) + \sum_{\{x,y\} \in \binom{N(v)}{2}} p_F(x,y) \geq c(K_k) + \binom{k}{2} Q_k,$$

using the induction hypothesis for $c(F)$, the path lemma for each summand, and $d_G(v) \geq k$ for the number of summands.

On the other hand, applying Lemma 4.1 to K_{k+1} at a vertex v shows that the cycles of K_{k+1} through v are in bijection with a pair $\{x, y\} \subseteq V(K_{k+1}) \setminus \{v\}$ together with an x - y path of K_k ; there are exactly $\binom{k}{2} Q_k$ of them. Hence

$$c(K_{k+1}) = c(K_k) + \binom{k}{2} Q_k,$$

which closes the induction.

For equality, first suppose $k \geq 4$. If $F \not\cong K_k$ then the induction hypothesis is strict, since $k-1 \geq 3$, and $c(G) > c(K_{k+1})$. If $F \cong K_k$ then G has $k+1$ vertices and, being k -connected on $k+1$ vertices, is K_{k+1} — which attains equality, as the displayed identity shows. So equality for $k \geq 4$ forces $G \cong K_{k+1}$.

It remains $k = 3$, where F is 2-connected and equality forces $c(F) = c(K_3) = 1$ as well as $d_G(v) = 3$. As in the proof of Corollary 4.3, $c(F) = 1$ forces every vertex of F to have degree 2 in F ; since $\delta(G) \geq 3$, each of them must be adjacent to v , so $d_G(v) = |V(F)| = n-1$. Together with $d_G(v) = 3$ this gives $n = 4$, and the only 3-connected graph on four vertices is K_4 . $\square$

Corollary 6.2. *Every 2-connected graph has at least $c(K_3) = 1$ cycle, with equality exactly for the cycles.*

Proof. A 2-connected graph contains a cycle. If $c(G) = 1$ then every vertex of G has degree 2, as argued above, so G is a cycle. $\square$

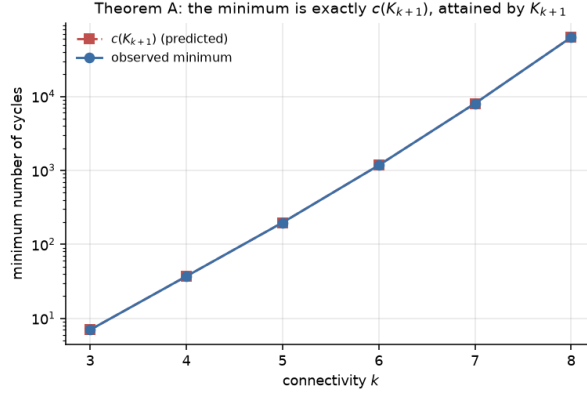


Figure 3: Theorem 6.1 on a logarithmic scale: the observed minimum over k -connected graphs equals $c(K_{k+1})$ for $3 \leq k \leq 8$, and is attained by K_{k+1} alone.

Theorem 6.1 settles the extremal question for each connectivity class, but not order by order. Combining it with Lemma 4.1 gives a family of explicit lower bounds, all proved here and all verified numerically in Section 8:

$$c(G) \geq \max \left\{ c(K_k) + \binom{\Delta(G)}{2} Q_k, \quad |E(G)| - |V(G)| + 1 \right\}, \quad k = \kappa(G). \quad (3)$$

The first term of (3) is the induction hypothesis of Theorem 6.1, refined by the degree of the deleted vertex. The second is the fundamental-cycle bound: the cotree edges of any spanning tree span pairwise distinct fundamental cycles, and k -connectivity gives $|E(G)| \geq k|V(G)|/2$.

7 Finiteness and decidability of the spectra

Theorem 7.1. *Let $k \geq 3$ and let G be a k -connected graph with $c(G) \leq K$, where $K \geq 1$. Then*

$$|V(G)| \leq \left\lfloor \frac{2(K-1)}{k-2} \right\rfloor.$$

In particular, for each K there are only finitely many k -connected graphs with at most K cycles.

Proof. Write $n = |V(G)|$. Since G is k -connected, $\delta(G) \geq k$, so $|E(G)| \geq kn/2$. Fundamental cycles of a spanning tree are pairwise distinct, so $K \geq c(G) \geq |E(G)| - n + 1 \geq (k-2)n/2 + 1$, which rearranges to the claim. $\square$

Corollary 7.2. *For every $k \geq 3$ the set S_k is a computable subset of the positive integers: for any m , the question “ $m \in S_k$?” is decidable by a finite search. Moreover $\min S_k = c(K_{k+1})$ by Theorem 6.1, and the function $\kappa(m) := \max\{\kappa(G) : c(G) = m\}$ is finite with*

$$m \geq \binom{\kappa(m)+1}{3}, \quad \text{so} \quad \kappa(m) = O(m^{1/3}).$$

Proof. By Theorem 7.1 a k -connected graph with $c(G) \leq m$ has at most $2(m-1)$ vertices and hence $k \leq 2(m-1)$, so $\kappa(m)$ exists and is finite. The bound $m \geq \binom{\kappa(m)+1}{3}$ follows from Theorem 6.1 together with the fact that K_{k+1} contains exactly $\binom{k+1}{3}$ triangles, all of which are cycles. For the computability assertion, for a given m enumerate all graphs on at most $2(m-1)$ vertices, test k -connectivity, and compute c for those that pass; the search terminates, and Theorem 7.1 guarantees that no witness for m is missed. $\square$

Table 2: Isomorphism classes produced by the generators, against the published cardinalities. The three-connectivity count at $n = 9$ is 80890.

n	2-connected			3-connected			brute force
	ours	published	ok	ours	published	ok	
3	1	1	yes	–	–	–	yes
4	3	3	yes	1	1	yes	yes
5	10	10	yes	3	3	yes	yes
6	56	56	yes	17	17	yes	yes
7	468	468	yes	136	136	yes	yes
8	7123	7123	yes	2388	2388	yes	–
9	194066	194066	yes	80890	80890	yes	–

The existence half is what makes Theorem 7.1 useful and also what makes it incomplete: the theorem reduces “ $m \in S_k$?” to a finite search but does not say how large a search is needed, and for S_3 the relevant threshold is unknown (Section 9).

No analogue of Theorem 7.1 can hold for $k \leq 2$: the cycle C_m has $c(C_m) = 1$ for every m , and the necklace of m triangles is a connected graph with exactly m cycles, so S_1 is the set of all positive integers while S_2 misses only the six exceptions of [1].

8 Computational results

8.1 The generators and how they are validated

Three exact primitives underlie the computations: cycle counting by depth-first search with the smallest vertex of the cycle as a pivot; vertex connectivity by exhaustive vertex-deletion tests; and exact isomorphism through canonical labelling (bliss, via **igraph** [16]) rather than heuristics such as colour refinement, which would not certify completeness.

The generator for 2-connected graphs uses the ear theorem: every 2-connected graph is a cycle, or is obtained from a 2-connected graph on fewer vertices by adding an ear with at least one fresh internal vertex, or from a 2-connected graph on the same number of vertices by adding a single edge (which preserves 2-connectivity). The generator for 3-connected graphs uses Menger: if G is 3-connected then $G - v$ is 2-connected for every v , so every 3-connected graph is obtained by adding one vertex to a 2-connected graph on one vertex fewer, joined to a set S of at least three vertices; and since $\delta(G) \geq 3$, S must moreover contain every degree-2 vertex of the parent, which is used as an exact pruning rule. Both generators are therefore complete, and both are deduplicated by canonical label.

Cardinality agreement alone would not certify completeness, so the generators are additionally checked against exhaustive brute force: for $n \leq 7$ we enumerate all $2^{\binom{n}{2}}$ graphs, filter by connectivity, and compare the resulting *sets of canonical labels* with those produced by the generators. For $n \leq 7$ the two agree exactly for both connectivity classes, and in particular the induced cycle-count spectra agree. We also checked c and $p(x, y)$ against **networkx** and against an independent brute-force search over 2-regular connected edge subsets.

8.2 Results

(i) **Theorem A.** Over all 3-connected graphs of order at most 9 (80890 classes) the minimum number of cycles at each connectivity level $3 \leq k \leq 8$ equals $c(K_{k+1})$, and the extremiser is K_{k+1} itself:

k	3	4	5	6	7	8
min c	7	37	197	1172	8018	62814

Figure 3 plots these against the predicted values.

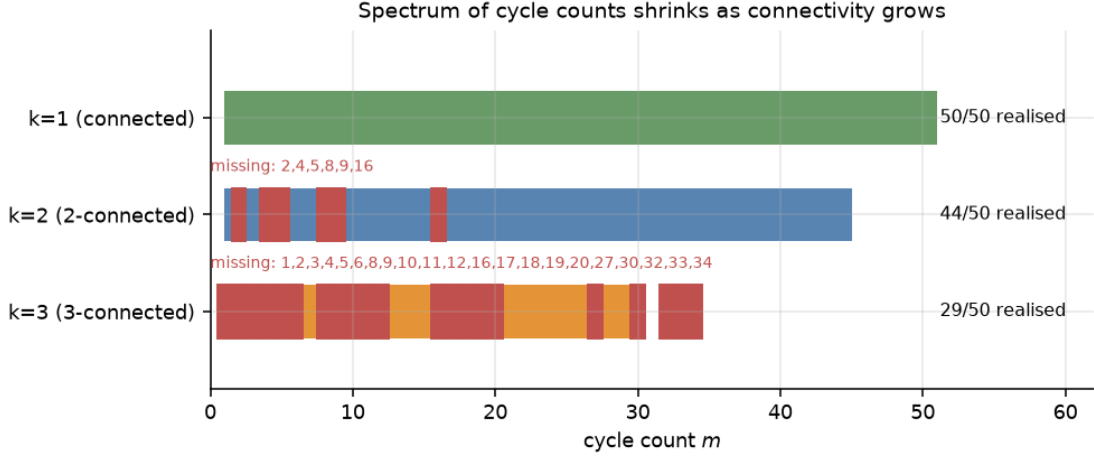


Figure 4: The spectra of S_1, S_2, S_3 below 51. Red marks missing values; the annotations give the exact missing sets.

(ii) **The spectra.** Recomputing the exception sets of S_2 and S_3 below 51 from our own generators gives:

- $S_1 = \mathbb{Z}_{>0}$: every $m \geq 1$ is realised by the necklace of m triangles, a connected graph with exactly m cycles.
- S_2 misses exactly $\{2, 4, 5, 8, 9, 16\}$ in $[1, 50]$, i.e. nothing beyond the exceptions proved by [1] (OEIS [15]).
- S_3 misses exactly $\{1, \dots, 6; 8, \dots, 12; 16, \dots, 20; 27; 30; 32; 33; 34\}$ in $[1, 50]$ — precisely the list conjectured in Section 6(a) of [1].

The last bullet is an independent confirmation of that conjecture in this range, obtained with a different generator; it is not a proof of the conjecture. Figure 4 displays the three spectra.

(iii) **The extremal function $m_3(n)$.** Exhaustive computation over all 3-connected graphs gives Table 3.

Table 3: $m_3(n)$ for $4 \leq n \leq 9$, with the minimisers. The last column is the conjectured lower bound of Conjecture 8.1.

n	$m_3(n)$	minimisers	degree sequence of a minimiser	$c(W_n)$	$\binom{n-1}{2}$
4	7	1	(3, 3, 3, 3)	7	3
5	13	1	(4, 3, 3, 3, 3)	13	6
6	14	1	(3, 3, 3, 3, 3, 3)	21	10
7	24	1	(4, 3, 3, 3, 3, 3, 3)	31	15
8	26	1	(3, 3, 3, 3, 3, 3, 3, 3)	43	21
9	42	2	(4, 3, 3, 3, 3, 3, 3, 3, 3)	57	28

The minimiser at $n = 4$ is K_4 ; at $n = 5$ it is the wheel W_5 ; at $n = 6$ it is the triangular prism $C_3 \times K_2$; at $n = 8$ it is a cubic graph. From $n = 6$ onwards wheels are *not* extremal, which is exactly why Theorem 6.1 does not determine $m_3(n)$ order by order.

(iv) **Conjecture C.** Every value of $m_3(n)$ in Table 3 respects $\binom{n-1}{2}$; we state this as a conjecture rather than a theorem (Section 9).

Conjecture 8.1. If G is a 3-connected graph on $n \geq 4$ vertices then $c(G) \geq \binom{n-1}{2}$. Among 3-connected graphs with a dominating vertex the minimiser is the wheel (this part is a theorem, Corollary 4.3).

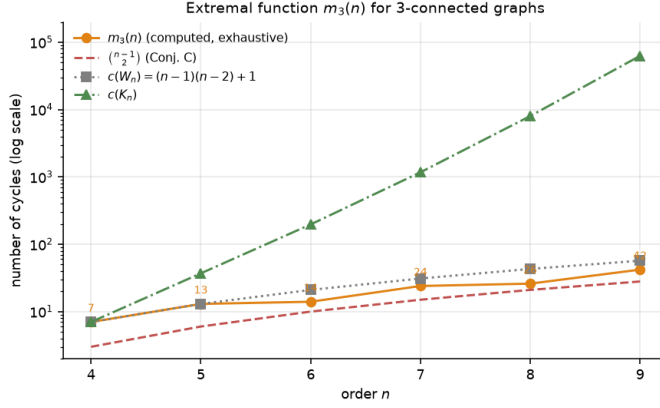


Figure 5: $m_3(n)$ against the wheel, the complete graph and the conjectured bound $\binom{n-1}{2}$ (logarithmic scale).

8.3 Complexity

All computations are exact integer computations; the following refers to the implementation in `src/cycle_spectra`.

- **Cycle counting.** Fixing the smallest vertex of the cycle as a pivot and enumerating simple paths from it, the running time is $O(\#\text{paths of } G)$ bitmask operations with $O(n)$ bits of working memory. Since $\#\text{paths} \leq \sum_{j \leq n} \binom{n}{j} (j-1)! = O((n-1)!)$ the worst case is factorial in n ; empirically the routine handles $n = 12$ in milliseconds. Counting c and $p(x, y)$ are the pipeline's bottlenecks.
- **Connectivity.** Testing k -connectivity by exhaustive deletion costs $O(\sum_{j < k} \binom{n}{j} (n + |E|))$, exponential in k and negligible at the sizes used here.
- **Generation.** The 2-connected generator performs $O(\sum_{j < n} \#2\text{-conn}(j) \binom{j}{2})$ candidate ear extensions, plus a closure under adding single edges; the number of canonical labellings is of the same order. Measured times appear in Figure 6; generating all 2-connected graphs on 9 vertices (194066 classes) takes about four minutes in pure Python, and the 3-connected generator on 9 vertices from the 2-connected generator on 8 takes about a minute. The 3-connected generator on 10 vertices did not finish within fifty minutes on the same machine.
- **Asymptotics.** The class counts of Table 1 have strictly increasing successive ratios (from 3 at $n = 3, 4$ to about 27 between $n = 8$ and $n = 9$), so enumeration is super-exponential in n by nature, not by weakness of the implementation. Nothing in the proofs depends on the enumeration.

9 Limitations and open problems

We are explicit about what this paper does not establish.

Conjecture 8.1 is open.

We verify it exhaustively for $n \leq 9$, and it holds for graphs with a dominating vertex (Corollary 4.3), but a cubic 3-connected graph of order 20 is out of reach of the bounds proved here: (3) gives only $c(G) \geq c(K_4) = 7$ and $c(G) \geq 3|E| - 3n + 1 = 2n - 1$. A proof needs a genuinely new way of trading the number of long paths against the number of long cycles. We also note that Conjecture 8.1 is loose on the computed range: $m_3(n)/\binom{n-1}{2}$ takes the values $7/3, 13/6, 14/10, 24/15, 26/21, 42/28$ for $n = 4, \dots, 9$.

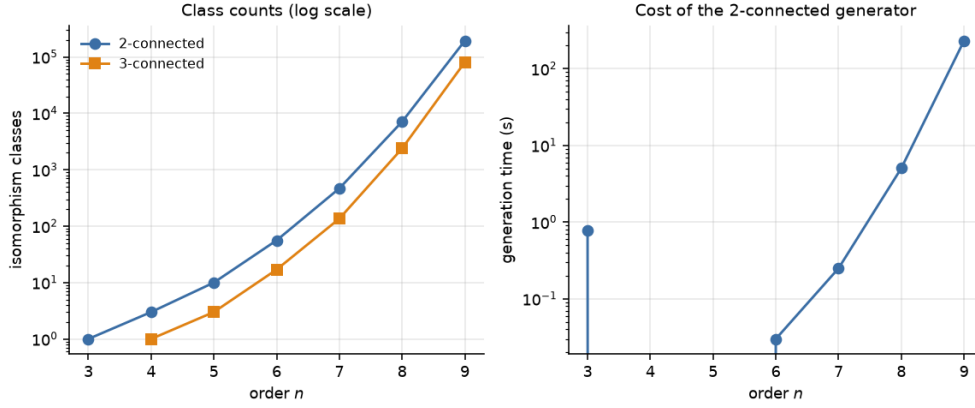


Figure 6: Left: growth of the enumerated class counts (logarithmic scale). Right: measured cost of the 2-connected generator.

Conjecture 6(a) of [1] is not settled.

Our computation agrees with it below 51, but ruling out a cycle count m for all n requires an order bound in terms of m that is (i) valid for all 3-connected graphs and (ii) small enough to be enumerable. Theorem 7.1 gives $|V(G)| \leq 2(m-1)$, far too weak, while Conjecture 8.1 would give $|V(G)| \leq 9$ for $m \leq 34$ — exactly the range we enumerate. There is therefore a clean logical dependency: the low part of the 3-connected spectrum is conjecturally one finite computation away.

$m_3(n)$ is known only for $n \leq 9$.

Our generators produce all 3-connected graphs of order at most 9 (80890 classes); $m_3(10)$ is not reported rather than guessed.

The spectra S_k for $k \geq 4$ are only sampled.

On orders at most 7 we find $|S_4| = 30$ and $|S_5| = 5$; Theorem 6.1 supplies the exact minima $c(K_5) = 37$ and $c(K_6) = 197$ for those classes, but the exception sets of S_4 are undetermined.

Equality in Lemma 5.2 for $r = 2$.

The bound $p_F(x, y) \geq Q_3 = 2$ is sharp, but equality is achieved by every 2-connected graph in which x and y are joined by exactly two paths, a large and structurally unclassified class containing all cycles. Theorem 6.1 avoids this case by using $\delta(G) \geq 3$ in the inductive step.

No claim about the existence direction of Conjecture 6(a).

Showing that every $m \geq 35$ is the cycle count of some 3-connected graph is a construction problem that we did not attack; our computation only certifies existence for $m \leq 34$.

Two directions seem promising. First, $m_3(n)$ appears to be attained by graphs that are as regular as their order allows — every minimiser we found is 3-regular when $3n$ is even, and otherwise has the most balanced degree sequence permitted by parity — and proving that heuristic would turn Table 3 from a computation into a theorem. Second, Theorem 6.1 says nothing about graphs of prescribed order; a sharp order-sensitive bound is exactly what Conjecture 8.1 asks for, and its failure to follow from the path-counting machinery suggests that the right object is not the number of paths but the number of cycles through a fixed vertex, where the extremal configuration is a cone.

10 Conclusion

Two clean statements organise the cycle content of highly connected graphs: among cones, wheels are extremal, and complete graphs are the sparsest k -connected graphs of their class (Theorems A

and B); and the spectra S_k are computable with an explicit order reduction (Theorem C). Computationally we obtain $m_3(n)$ for $4 \leq n \leq 9$ and reproduce the conjectured exception set of S_3 below 51. The open core is Conjecture 8.1, the order-sensitive version of Theorem A, which we have identified as the single missing ingredient for turning the low part of the 3-connected spectrum into a finite computation.

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A Identities

A.1 The cycles of a complete graph

A cycle of length j in K_n is a j -element vertex set together with one of $(j-1)!/2$ cyclic orders, so

$$c(K_n) = \frac{1}{2} \sum_{j=3}^n \binom{n}{j} (j-1)!,$$

giving 1, 7, 37, 197, 1172, 8018, 62814 for $n = 3, \dots, 9$, in agreement with [14] and with the values computed independently by our code.

A.2 The recurrence for Q_k

Recall $Q_m = \sum_{j=0}^{m-2} \binom{m-2}{j} j!$. Since $\binom{m-2}{j} j! = (m-2)!/(m-2-j)!$, putting $E_t = \sum_{i=0}^t 1/i!$ and substituting $i = m-2-j$ gives $Q_m = (m-2)! E_{m-2}$, whence for $r \geq 2$

$$Q_{r+1} = (r-1)! E_{r-1} = (r-1)! \left(E_{r-2} + \frac{1}{(r-1)!} \right) = (r-1)(r-2)! E_{r-2} + 1 = (r-1)Q_r + 1,$$

and $Q_2 = 0! E_0 = 1$. This proves Lemma 5.1.

A.3 Why $c(F) = 1$ forces F to be a cycle

If $c(F) = 1$ and some vertex x has degree $d \geq 3$, pick distinct neighbours y, z of x . In a 2-connected graph there is a y - z path avoiding x (otherwise x is a cut vertex), so $xyPzx$ is a second cycle, a contradiction. Hence $\delta(F) = 2$ whenever $c(F) = 1$, and a 2-regular graph is a disjoint union of cycles, of which only one may be present since $c(F) = 1$. Symmetrically, $c(K_1 \vee F) = 1$ is impossible for $|V(F)| \geq 3$.

B What the accompanying code checks

- c and $p(x, y)$ agree with `networkx` on all graphs with at most five vertices, and with an independent brute-force search over 2-regular connected edge subsets on all graphs with at most five vertices; the $n = 6$ case is checked by brute force as well.
- κ agrees with `networkx` on all graphs with at most six vertices.
- The generators agree with exhaustive brute force, as *sets of canonical labels*, for both connectivity classes and $n \leq 6$ inside the test suite, and for $n \leq 7$ via `scripts/validate_generators.py`.
- Lemma 5.2 holds on all pairs of vertices of all 2-connected graphs with $n \leq 7$, with its equality condition checked exhaustively for $r \geq 3$ and a recorded counterexample to the naive statement at $r = 2$.
- Theorem 6.1 and Corollary 4.3 hold on all 3-connected graphs with $n \leq 9$.
- Conjecture 8.1 holds on all 3-connected graphs with $n \leq 7$.
- The spectrum statements of Section 8(ii) are regression tested.